

WHEN REPRESENTATION HEALTH DOESN'T GUARANTEE PERFORMANCE: PITFALLS IN ADAPTIVE DATA MIXING FOR FOUNDATION MODELS

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ABSTRACT

Foundation models are typically pre-trained on diverse data sources, but determining optimal mixing ratios remains challenging. We investigate whether monitoring geometric properties of learned representations—attention entropy Vaswani et al. (2017), feature space clustering quality Rousseeuw (1987), and layer-wise similarity Kornblith et al. (2019)—can guide dynamic data mixing during pre-training. While representation-guided mixing successfully reduces cross-domain performance imbalance (4.4% improvement in standard deviation), it paradoxically degrades validation loss by 5.2% and reduces average domain accuracy. Our experiments on transformer models trained across six diverse domains reveal a fundamental tension: optimizing for representation health metrics does not align with optimizing for generalization. We find that interventions triggered by representation degradation often harm performance on well-learned domains without sufficient gains on struggling domains. This work highlights a critical pitfall in data-centric optimization and demonstrates that geometric signatures alone are insufficient for guiding adaptive data mixing without careful consideration of downstream objectives.

1 INTRODUCTION

Foundation models achieve remarkable capabilities by training on diverse data sources spanning code, scientific literature, conversational text, and general knowledge Goodfellow et al. (2016). However, determining optimal mixing ratios across these sources remains an open challenge. Static mixing strategies often lead to imbalanced learning where models excel on some domains while underperforming on others. Recent work has proposed adaptive data mixing methods that dynamically adjust sampling weights during training Li et al. (2025); Corrado et al. (2025). These approaches typically rely on task-specific loss signals or gradient conflicts to guide mixing decisions. However, such methods are inherently reactive—they respond after performance issues manifest in loss or gradient statistics.

We investigate an alternative hypothesis: can intrinsic properties of learned representations serve as early indicators of data source imbalance? Specifically, we monitor three geometric signatures during pre-training: (1) attention entropy across transformer layers Vaswani et al. (2017), measuring specialization patterns; (2) silhouette scores Rousseeuw (1987), quantifying feature space clustering quality; and (3) layer-wise CKA similarity Kornblith et al. (2019), detecting premature convergence. Our experiments reveal a surprising negative result: while representation-guided mixing successfully improves balance metrics (reducing cross-domain performance variance by 4.4%), it consistently degrades both validation loss (5.2% increase) and average domain performance. This finding exposes a fundamental pitfall in using representation geometry to guide data mixing—optimizing for representation health does not guarantee improved generalization.

2 RELATED WORK

Adaptive Data Mixing. Recent methods like PiKE Li et al. (2025) optimize sampling weights based on gradient conflicts between tasks, while AutoMixAlign Corrado et al. (2025) uses minimax

optimization to balance performance across domains. These approaches are reactive, responding to degradation signals in loss or gradients. Our work differs by attempting to use proactive representation health monitoring, though our results suggest this approach faces significant challenges. **Representation Analysis.** Prior work has studied attention patterns Vaswani et al. (2017), clustering quality Rousseeuw (1987), and representation similarity Kornblith et al. (2019) as diagnostic tools. However, no prior work has attempted to leverage these metrics for guiding data source selection during pre-training. Our negative results suggest that while these metrics provide interpretable insights, they may not be suitable optimization targets for data mixing.

3 METHOD

Representation Health Metrics. We monitor three complementary metrics computed every N training steps: (1) *Attention Entropy*: mean entropy of attention weights across heads and layers, where lower values indicate over-specialization; (2) *Silhouette Score*: clustering quality of domain-labeled embeddings, measuring representation separation; (3) *CKA Similarity*: layer-wise representation similarity, where high values may indicate premature convergence. **Adaptive Mixing Controller.** Our controller establishes baseline metrics during the first three epochs, then monitors for degradation signals. Interventions are triggered when multiple concurrent signals indicate unhealthy dynamics: significant entropy drops ($> 20\%$ below baseline), persistent silhouette decline (three consecutive epochs below 85% of baseline), high CKA (> 0.90), or gradient imbalance across domains. When triggered, the controller adjusts sampling probabilities based on per-domain performance and gradient norms, with stronger adjustments during a critical window (epochs 5–10) where representation metrics change most rapidly.

4 EXPERIMENTAL SETUP

Model and Data. We train transformer models (128M parameters, 3 layers, 4 attention heads) on six diverse domains: Wikipedia (general knowledge), GitHub (code), arXiv (scientific), conversational text, TinyStories, and OpenWebText. Each domain contains 400 training and 100 validation samples with 32-token sequences. **Baselines and Evaluation.** We compare static uniform mixing against our representation-guided approach across 20 epochs with learning rate $1e-3$ and batch size 64. We evaluate using domain-specific next-token prediction accuracy and compute a balance score as $1/(1 + CV)$ where CV is the coefficient of variation across domain accuracies. We also measure validation loss, representation health metrics, and per-domain gradient norms.

5 RESULTS

The Balance-Performance Trade-off. Figure 1 reveals the central negative result. Validation loss (top-right) shows representation-guided mixing underperforms static mixing after epoch 10, achieving 3.566 versus 3.389—a 5.2% degradation. This occurs despite reducing domain imbalance from 0.218 to 0.208 std (4.4% improvement, bottom-left). This tension suggests interventions re-allocate capacity away from well-learned domains without sufficient gains on struggling domains. Per-domain accuracy (bottom-right) shows marginal improvements on low-performing domains like Wikipedia while maintaining similar performance on others, except notable OpenWebText degradation.

Domain-Specific Failure Modes. Figure 2 reveals why representation-guided mixing fails. On OpenWebText (left), the adaptive method plateaus around epoch 7–8 at 0.32 accuracy while static mixing continues to 0.375—a 17% gap. This divergence begins during the critical window, suggesting interventions incorrectly reduced sampling based on misaligned signals. In contrast, the code domain (right, representative of five others) shows both methods tracking identically, achieving similar final accuracies around 0.14. This pattern—close tracking on five domains but substantial degradation on one—explains why validation loss increases despite improved balance.

Representation Health & Interventions. Representation-guided mixing achieved slightly better final representation health: silhouette scores of 0.317 vs. 0.290, and lower attention entropy. However, these improvements did not translate to better validation loss or accuracy, evidencing that representation health optimizes objectives orthogonal to generalization (see Appendix Fig. 3). The

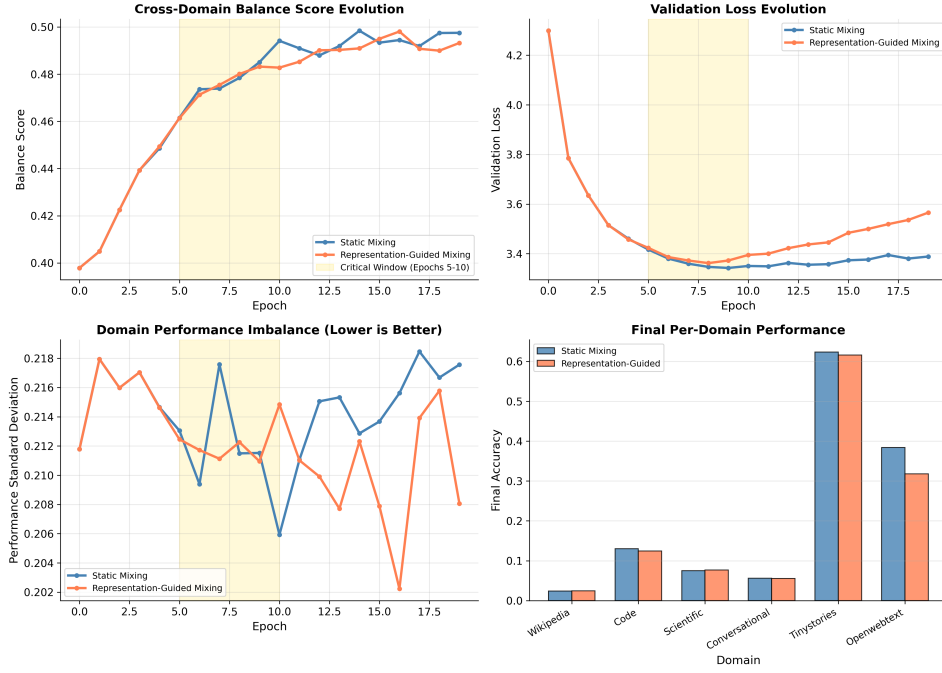


Figure 1: **Core Performance Trade-off.** Validation loss (top-right) shows representation-guided mixing (orange) degrades to 3.566 vs. static mixing’s 3.389 (5.2% increase). Domain imbalance (bottom-left) improves from 0.218 to 0.208 std (4.4% reduction). Critical window (epochs 5–10, shaded) marks rapid metric changes.

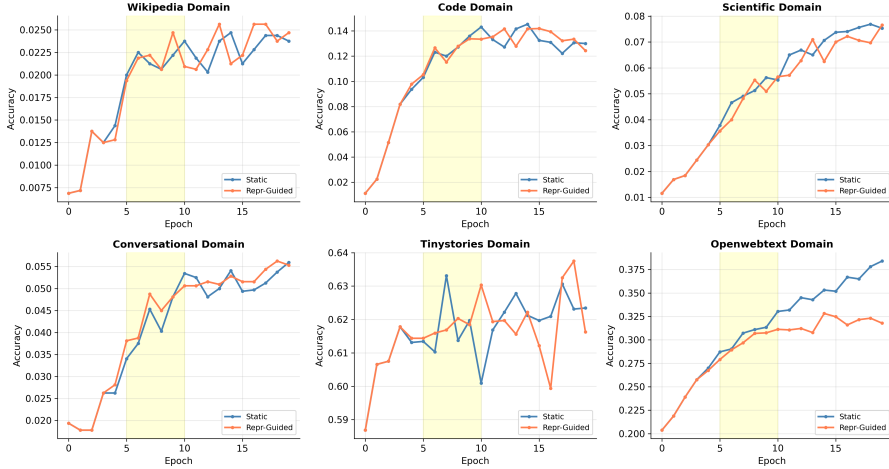


Figure 2: **Domain-Specific Learning Dynamics.** Left: OpenWebText shows substantial divergence where representation-guided mixing (orange) plateaus at 0.32 while static (blue) reaches 0.375. Right: Code domain (representative of five others) shows both methods tracking closely with similar final performance around 0.14.

controller triggered 9 interventions (3 during critical window, 6 after), yet validation loss remained consistently higher after epoch 10, suggesting mixing adjustments improved balance metrics but harmed held-out performance (see Appendix Fig. 4).

Ablation Studies. We examined intervention triggers, loss functions, curriculum strategies, and optimizers (Appendix Sec. C). Key findings: (1) Multi-metric triggering, entropy-only, and sequential priority achieved similar balance scores around 0.50, showing minimal trigger impact; (2) Domain-

contrastive losses improved silhouette scores 9.5% but provided no validation benefits, confirming the representation-generalization disconnect; (3) Curriculum strategies underperformed static mixing on both metrics; (4) AdamW substantially outperformed SGD and RMSprop (0.50 vs 0.41–0.43 balance). No design choice resolved the fundamental balance-performance trade-off, suggesting the issue is inherent to using representation health as an optimization target.

6 DISCUSSION AND LIMITATIONS

Our investigation reveals a critical pitfall: optimizing for geometric health metrics does not guarantee improved generalization. While our approach reduces cross-domain imbalance, it does so by reallocating capacity away from well-learned domains without sufficient compensatory gains. Several factors contribute: First, representation health metrics may capture intermediate dynamics orthogonal to final performance. Second, our controller lacks mechanisms to constrain interventions when they risk harming overall generalization. Third, the balance metric may not align with practical objectives where some domains naturally require more capacity.

These negative results provide valuable lessons. Future work should explore multi-objective optimization frameworks explicitly trading off balance against validation loss, incorporate held-out validation signals into mixing decisions, and develop domain-specific capacity allocation rather than assuming equal importance. Our findings underscore the importance of validating that intermediate optimization targets align with downstream objectives before deploying adaptive training strategies.

REFERENCES

- Nicholas E. Corrado, Julian Katz-Samuels, Adithya M. Devraj, Hyokun Yun, Chao Zhang, Yi Xu, Yi Pan, Bing Yin, and Trishul M. Chilimbi. Automixalign: Adaptive data mixing for multi-task preference optimization in llms. *ArXiv*, abs/2506.00569, 2025.
- Ian Goodfellow, Yoshua Bengio, Aaron Courville, and Yoshua Bengio. *Deep learning*, volume 1. MIT Press, 2016.
- Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey E. Hinton. Similarity of neural network representations revisited. *ArXiv*, abs/1905.00414, 2019.
- Zeman Li, Yuan Deng, Peilin Zhong, Meisam Razaviyayn, and V. Mirrokni. Pike: Adaptive data mixing for large-scale multi-task learning under low gradient conflicts. 2025.
- P. Rousseeuw. Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65, 1987.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and I. Polosukhin. Attention is all you need. pp. 5998–6008, 2017.

SUPPLEMENTARY MATERIAL

A COMPLETE REPRESENTATION HEALTH METRICS

B DETAILED INTERVENTION ANALYSIS

C ADDITIONAL ABLATION STUDIES

D INDIVIDUAL ABLATION RESULTS

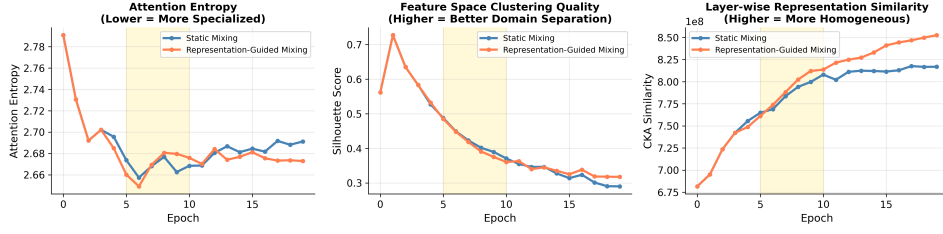


Figure 3: **Representation Health Evolution.** Three panels showing (left) attention entropy, (middle) silhouette score for feature space clustering, and (right) CKA similarity. All metrics show substantial changes during the critical window (epochs 0–5, shaded), but representation-guided mixing (orange) does not achieve substantially better final values than static (blue). Final values: silhouette 0.317 vs 0.290, entropy slightly lower for adaptive, CKA converging around $8.0\text{--}8.25 \times 10^8$ for both.

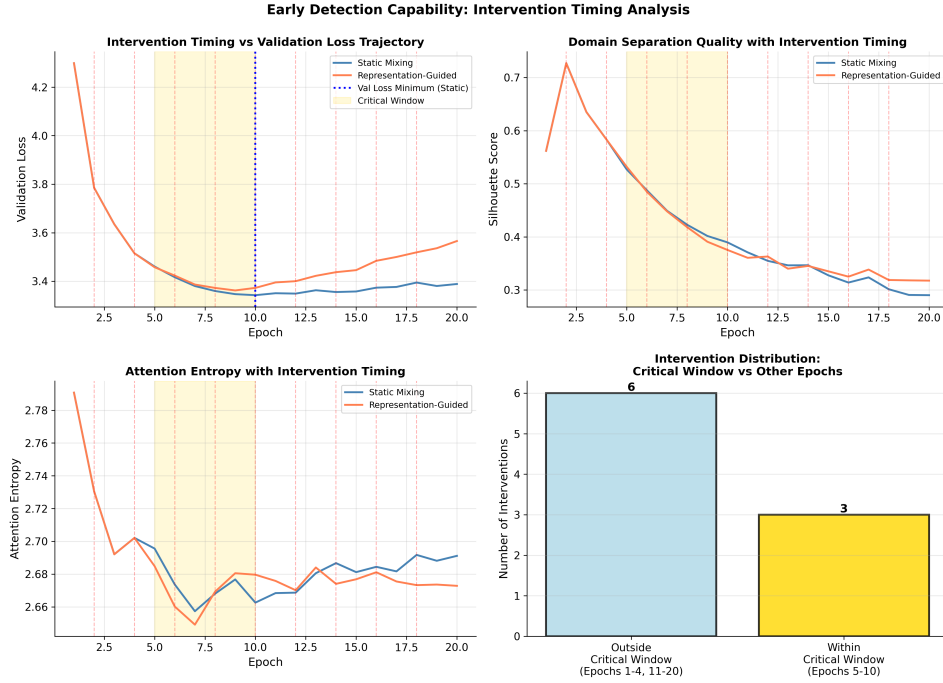


Figure 4: **Intervention Timing Analysis.** Top-left: Validation loss with vertical red lines marking 9 interventions. Top-right: Domain separation via silhouette score. Bottom-left: Attention entropy evolution. Bottom-right: Intervention distribution showing 6 outside and 3 within critical window (epochs 5–10, shaded). Despite interventions, validation loss for representation-guided (blue) remains higher than static (orange).

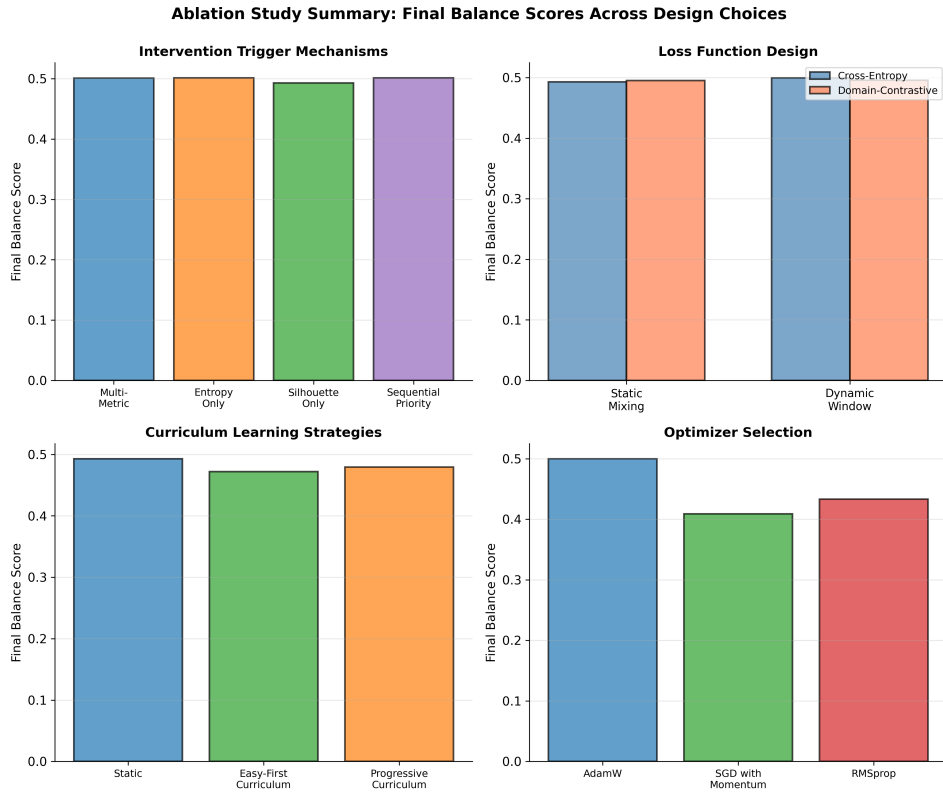


Figure 5: **Comprehensive Ablation Summary.** Final balance scores across four design dimensions. Multi-metric triggering, entropy-only, and sequential priority achieve similar scores around 0.50. Domain-contrastive losses show no advantage over cross-entropy. Curriculum strategies underperform static mixing. AdamW substantially outperforms SGD and RMSprop (0.50 vs 0.41–0.43).

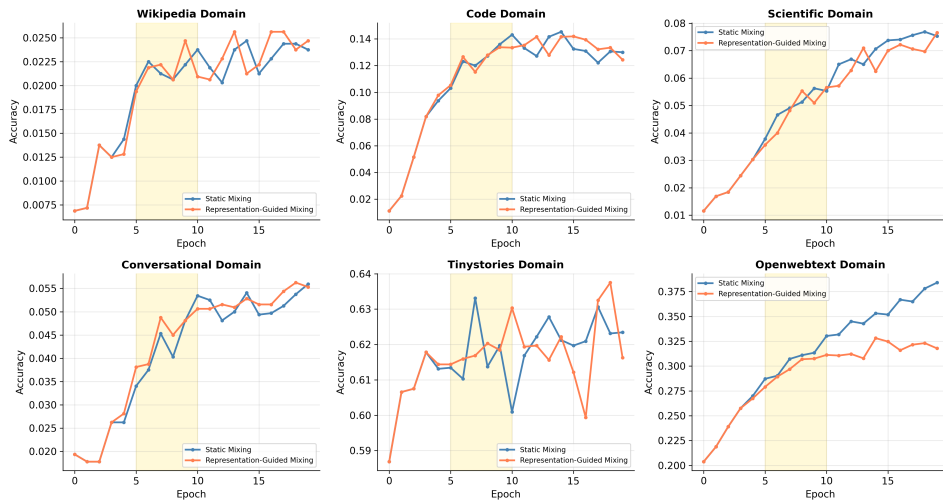


Figure 6: **Complete Per-Domain Training Dynamics.** Performance trajectories across all six domains showing representation-guided (orange) tracks static (blue) closely on five domains but diverges substantially on OpenWebText (bottom-right), plateauing around epoch 7–8 at 0.32 while static continues to 0.375. Yellow shading indicates critical window (epochs 5–10).

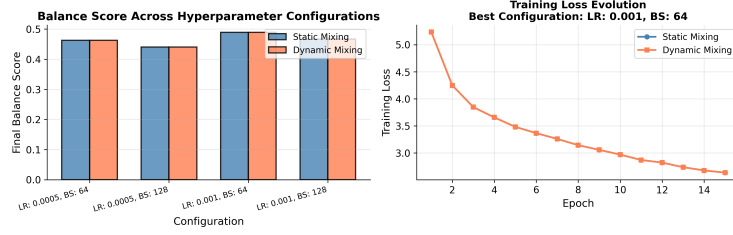


Figure 7: **Baseline Hyperparameter Analysis.** Balance scores vary more with learning rate and batch size (0.44–0.49) than with mixing strategy. Best configuration (LR=1e-3, BS=64) achieves 0.49 for both static and dynamic mixing, while worst (LR=5e-4, BS=128) achieves 0.44. Left: balance scores across configurations; right: training loss evolution for best configuration.

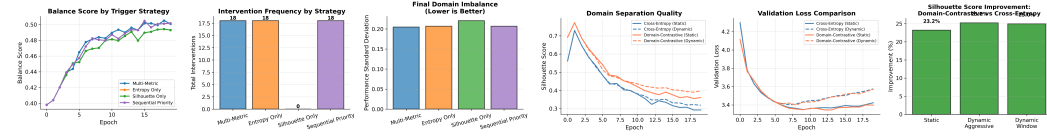


Figure 8: **Individual Ablations.** Left: Balance scores across different intervention trigger mechanisms showing minimal variation. Right: Comparison of cross-entropy vs. domain-contrastive loss functions under static and dynamic mixing, all achieving similar performance.

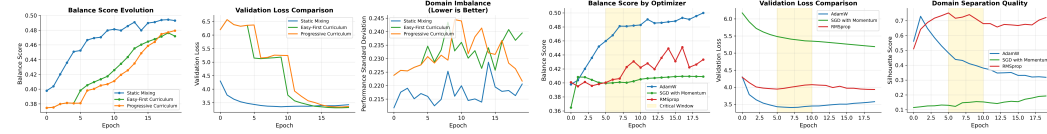


Figure 9: **Additional Individual Ablations.** Left: Curriculum learning strategies (easy-first and progressive) underperform static mixing. Right: Optimizer comparison showing AdamW substantially outperforming SGD with momentum and RMSprop.

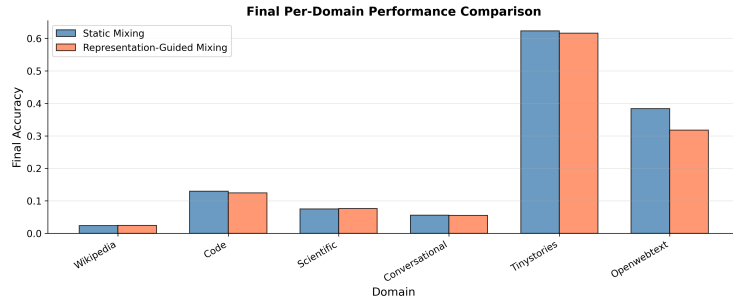


Figure 10: **Final Domain Performance Comparison.** Per-domain accuracy showing representation-guided mixing achieves comparable or slightly worse performance across all six domains compared to static mixing, with largest gap on OpenWebText.